CLASS AGNATHA

Characteristics of Class Agnatha

- These are vertebrates without jaws. This group flourished in the middle Paleozoic era and was represented by several orders collectively known as Ostracoderms (meaning shell skins).
- Surviving members of this class are the hagfishes and lampreys.

The main distinguishing characteristics of these animals are;

- ✓ They **lack jaws**
- ✓ They **lack paired fins**
- ✓ **Cartilaginous Skeleton:** They lack a bony skeleton; instead, their internal support structure is made entirely of flexible cartilage.
- ✓ **Circular, Suctorial Mouth:** They possess a round, funnel-like mouth often equipped with horny teeth used for scavenging or parasitism (especially in lampreys).
- ✓ **Smooth, Scale-less Skin:** Their skin is soft, glandular, and lacks the scales found in most other fish groups. Hagfish, in particular, are known for producing large amounts of protective slime.
- ✓ **Pore-like Gill Openings:** Instead of a single operculum (gill cover), they have a series of 7 or more external gill slits or pores along the sides of the head for respiration.
- ✓ **Persistent Notochord:** The notochord remains throughout the organism's life as the primary axial support, rather than being fully replaced by a vertebral column. So notochord persists in adults as the supporting structure.

Ostracoderms: These are extinct members of class Agnatha. Ostracoderms were small, fresh water and fishlike. They were dorso-ventrally flattened especially at the front of the body.

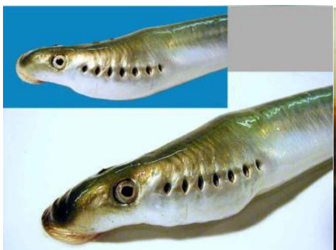
They had a single median nostril and pineal eye on top of the head. *The role of the pineal eye was to adjust the rate of activity to changing conditions of illumination.*

They had a medial\ fins, dorsal, ventral, caudal, but no paired fins.

The tail was **heterocercal;** the upper portion was larger and more ridged than the lower. Most of these characteristics are found in living agnathans.

Lampreys

- They have an elongate body that resembles that of eels.
- A sucker surrounds the mouth.
- The sucker is round when spread open for use but its lateral margins are drawn together when the lamprey swims.
- From the floor of the mouth a protrusible tongue projects into the mouth. A long central cartilage stiffens the tongue.
- Smaller cartilages at the tip of the tongue bear horny teeth that are used to rasp wounds in prey. Both sets of teeth are ectodermal, that is, they develop from the out layer of the embryo and not homologous to the teeth of higher vertebrates.
- When a lamprey is feeding blood is passed from the sucker into the gut without detaching the sucker. The sucker helps them attach to other fishes since they are parasites.
- Behind the sucker are two eyes without eyelids.
- On the dorsal surface of the head is a single nostril that is immediately followed by a pineal eye.
- Posterior to each eye are seven small openings, the gill slits.



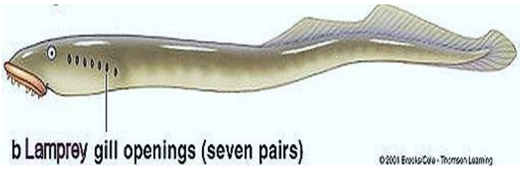
Hagfishes

- These are found in temperate seas.
- Unlike lampreys they are not parasitic so they do not have a sucker.
- They feed on small invertebrates and scavenge dead fish.
- They have teeth on the tongue that they use to scrap tissues from carcasses.
- Their body has small openings along the side of the body that lead to mucus secreting glands. The mucus is viscous and sticky, and protects the hagfish from predators.
- Their body fluids have the same ion concentration as seawater and so do not need special osmoregulatory adaptations.
- They retain in adults two types of kidneys, pronephros and mesonephros (***Pronephros*** *is the primitive, short-lived embryonic kidney, while* ***Mesonephros*** *is the temporary but functional embryonic kidney that bridges development before the permanent metanephros forms.*).

Hagfish slime

Distinguishing Features of Petromyzon and Myxine

Petromyzon	Myxine
1) It is commonly called **lamprey**	1) It is commonly called **hag fish** or **slime eel**
2) Internal ear has two semi-circular canals	2) Internal ear has one semi-circular canal
3) Cranial nerves are ten pairs	3) Cranial nerves are eight pairs
4) Seven pairs of gills are present	4) Six pairs of gills are present
5) It is a sanguivorous ectoparasite	5) It is a necrophagus animals
6) It does not produce slime	6) It produces enarmous amount of slime
7) Development is indirect (**Ammocoetus larva**)	7) Development is direct
8) Exhibits anadromous migration	8) Does not show anadramous migration



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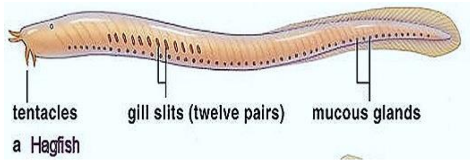


(b)

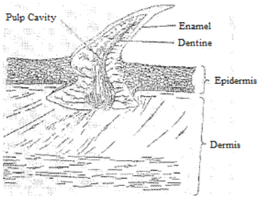
CLASS CHONDRICHTHYES (Cartilaginous fishes)

- They are the first group of vertebrates with jaws (perhaps one of the reasons sharks are so predacious).
- The development of jaws enabled vertebrates to diversify their mode of life and perhaps one of the reasons they out- competed the ostracoderms.
- These are living fishes whose skeleton is cartilaginous.
- The earliest known species were marine inhabitants and the group has remained primarily marine ever since although few have secondarily adapted to fresh water.
- Examples of cartilaginous fishes are dogfishes, shark, skates and rays.

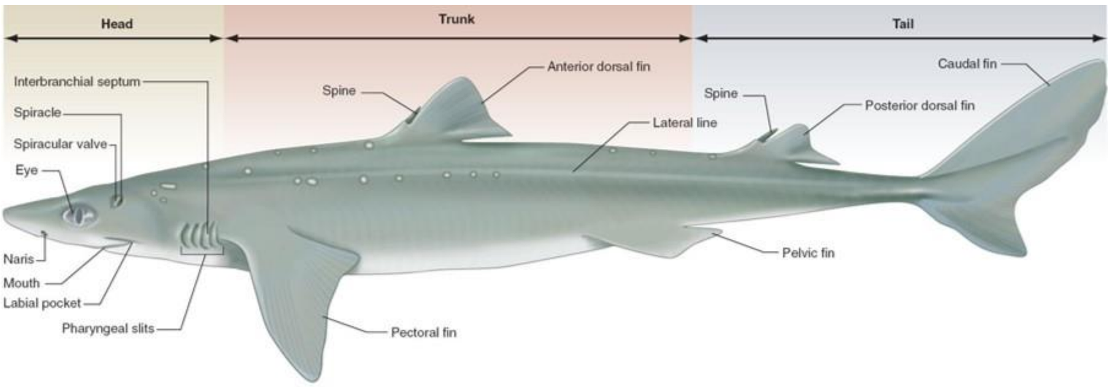
Characteristics of Chondrichthyes

- They have jaws
- They have a well-developed electro-receptive system.
- They do not have Bone scales are either tiny placoid scales or no scales.
- Internal skeleton is entirely cartilaginous
- Tail is heterocercal
- They have five pairs of gill pouches that open to the outside individually on the body surface.
- Males have a clasper on the pelvic fins with which sperms are transferred to the female.
- Fertilization is internal.
- Sense organs are well developed.

- Electro-reception plays a prominent role in feeding behaviour. The electroreceptors are ampullaty organs.
- They lie at the ends of long tubes filled with gelatinous material. These tubes are distributed all over the head in such a way that the fish can detect the source of weak electric currents generated by the muscle activities of prey organisms.



Placoid scales



External features of a cartilaginous fish.

Adaptive Radiation of Cartilaginous Fishes

Cartilaginous fishes are grouped into two subclasses:

Holocephalia,

- they have four gill slits that do not open directly to the outside but into a common chamber covered by a flap of skin.
  - The spiracle is absent.
  - Their upper jaw is fused to the cranium. The solid jaw allows them to feed on mollusks, crustaceans and other hard -bodied animals.
  - They have no scales.
  - Males have club-shaped clasping organ in addition to the pelvic claspers.


Chimaeras Ratfish

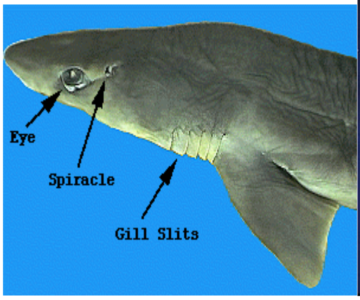
Elasmobranchii:

- They have five to seven gill slits without covers.
- Usually they have a pair of gill openings modified to spiracles.
- The upper jaw is movable.
- Placoid scales usually present.
- The subclass is divided into three groups: Squalomorpha, Galeomorpha and Batoidea.

Squalomorpha sharks are deep- water species. The dogfish belongs to this group.

Galeomorpha are subtropical and include the more familiar, predacious sharks "that are known to attack humans.

Batoidea are the skates and rays. Skates and rays are bottom dwellers and are dorsal-ventrally flattened. Trunk and tail muscles are considerably reduced and they swim along the bottom by undulations of their enormous pectoral fins. Many rays and skates exhibit interesting offensive and protective adaptations, among them snouts extended into "saws" in the saw-fishes, poison spines at the end of tail in sting rays and electric organs lateral to the eyes in torpedo rays or electric ray.



CLASS OSTEICHTHYES- BONY FISHES

Class Osteichthyes, or bony fishes, represents the largest class of vertebrates, accounting for over 96% of all fish species, comprising more than 28,000 species inhabiting both fresh and salt water. Characterized by a bony skeleton, gills covered by a bony plate called an operculum, and a swim bladder for buoyancy, this group ranges from tiny seahorses to large tuna.

General Features of Bony Fishes

- **Skeletal Structure**

Ossified Skeleton: Unlike Chondrichthyes (sharks and rays), their endoskeleton is primarily made of calcium-rich bone.

Skull: They possess a stable pattern of cranial bones and a rooted dentition (medial insertion of teeth).

- **External Anatomy**

Scales: The body is typically covered in **cycloid, ctenoid, or ganoid scales**, which are dermal in origin. Some species may be scaleless.

Fins: They possess both paired (pectoral and pelvic) and unpaired (dorsal, anal, and caudal) fins. These fins are supported by bony rays or spines.

Operculum: A bony flap called the **operculum** covers the gills on each side, allowing the fish to pump water over the gills even while stationary.

- **Buoyancy and Respiration**

Swim Bladder: Most bony fish possess a gas-filled organ known as a **swim bladder**. This helps the fish maintain neutral buoyancy, allowing it to "hover" at specific depths without wasting energy swimming.

Gills: Respiration occurs through four pairs of gills located in the pharyngeal pouches.

● Physiology and Sensory Systems

Lateral Line System: They have a well-developed lateral line system that detects vibrations and water pressure changes, aiding in navigation and prey detection.

Heart: The heart is two-chambered (one atrium and one ventricle), and they maintain a single-loop circulatory system.

Cold-Blooded: They are **ectothermic**, meaning their body temperature fluctuates with the surrounding environment.

● Reproduction and Development

Fertilization: In the majority of species, fertilization is **external** (oviparous), with females laying eggs and males fertilizing them in the water.

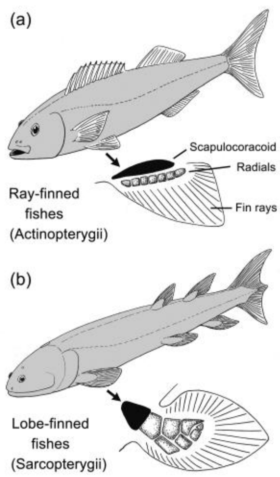
Development: Development is usually direct, though many species go through distinct larval stages.

● Classification

Osteichthyes is further divided into two major groups:

Actinopterygii (Ray-finned fishes): This includes the vast majority of familiar fish (e.g., Tilapia, Salmon, Goldfish). Their fins are webs of skin supported by bony spines.

Sarcopterygii (Lobe-finned fishes): A much smaller group including lungfish and coelacanths. Their fins are fleshy and lobed, joined to the body by a single bone, which is an evolutionary precursor to tetrapod limbs.



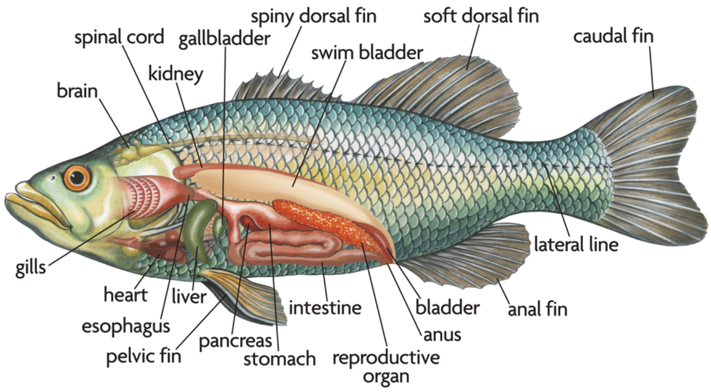
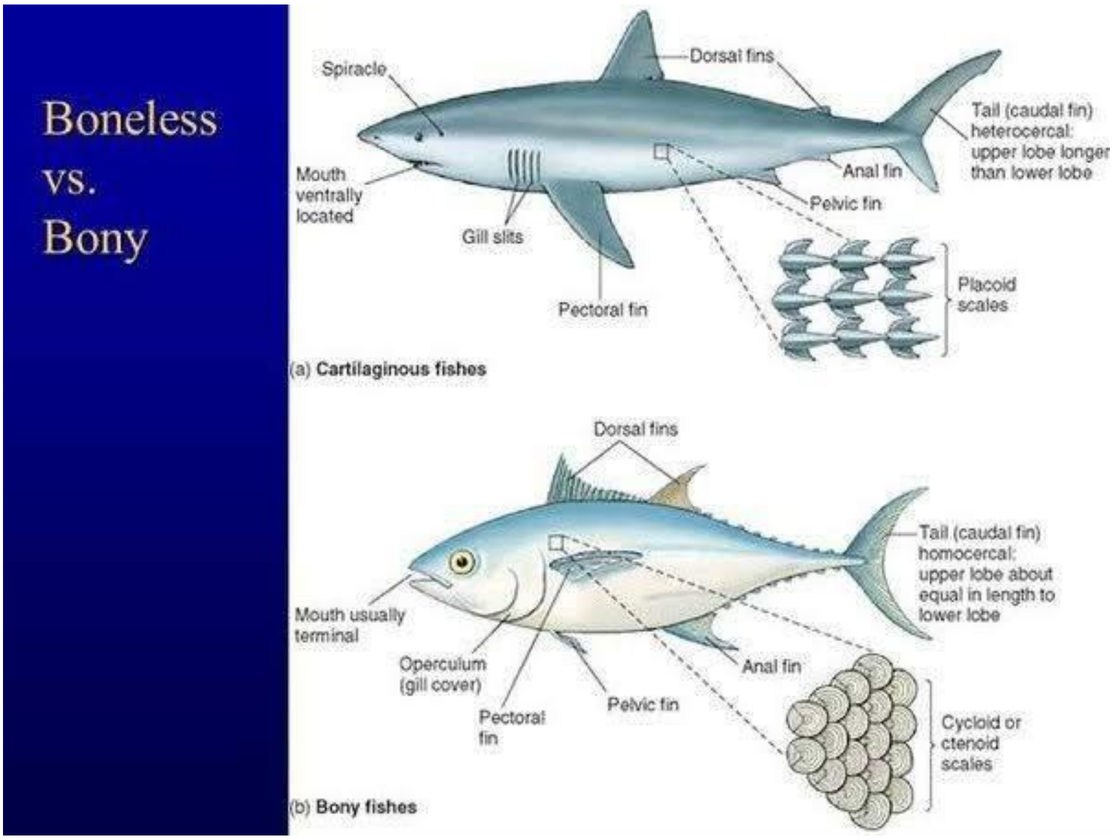


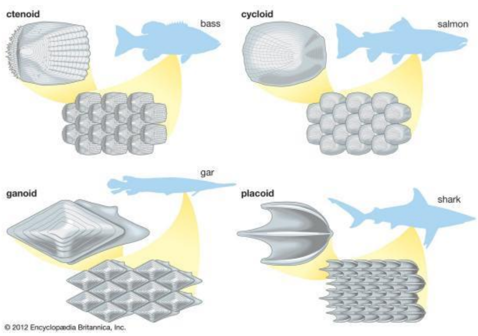
Table Showing Differences Between Osteichthyes and Chondrichthyes

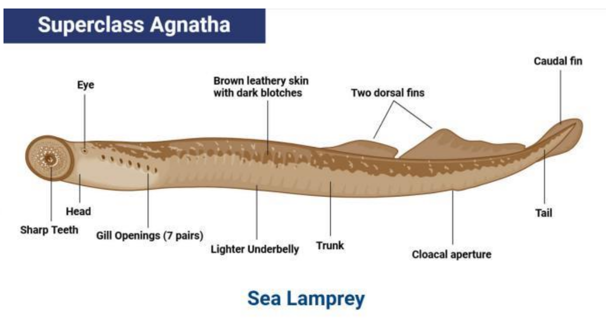
Feature	Osteichthyes (Bony Fish)	Chondrichthyes (Cartilaginous Fish)
Skeleton	Made of bone (calcium phosphate)	Made of cartilage (lighter, flexible)
Scales	Ganoid, cycloid, or ctenoid (thin, flexible)	Placoid scales (tooth-like, reduce friction)
Gill Structure	4 pairs of gills covered by operculum (bony flap)	5–7 pairs of gill slits, no operculum
Buoyancy	Swim bladder present for buoyancy	No swim bladder; buoyancy via oil-filled liver and fin movement
Mouth Position	Terminal (front of head)	Ventral (underside of head)
Tail (Caudal Fin)	Homocercal (upper and lower lobes equal)	Heterocercal (upper lobe larger)
Reproduction	Mostly external fertilization (eggs in water)	Mostly internal fertilization; oviparous or viviparous
Excretion	Ammonia is main nitrogenous waste	Urea is main nitrogenous waste
Habitat	Both freshwater and marine	Mostly marine
Examples	Tilapia, salmon, trout, lungfish	Sharks, rays, skates,

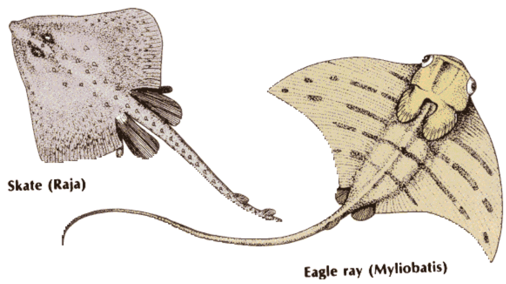
Pictorial Slide Showing Differences Between Osteichthyes and Chondrichthyes

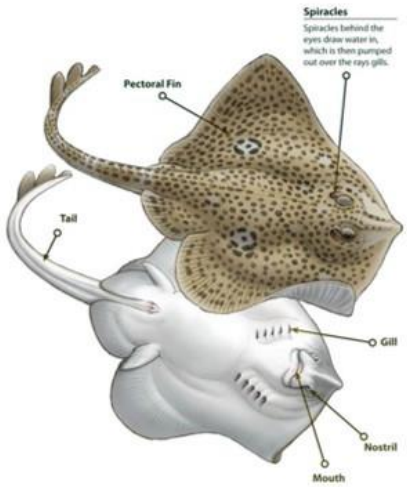


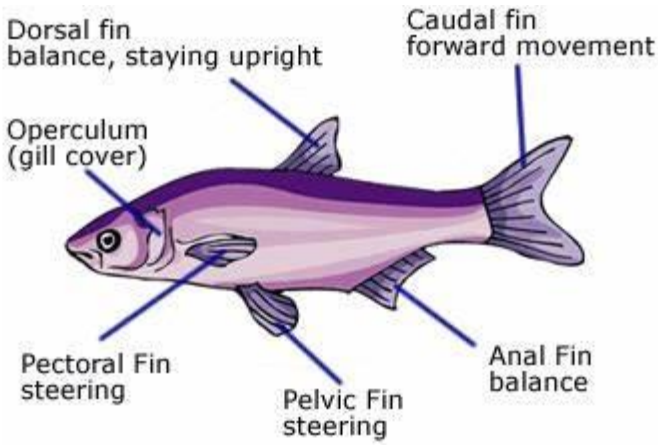
More Pictorials





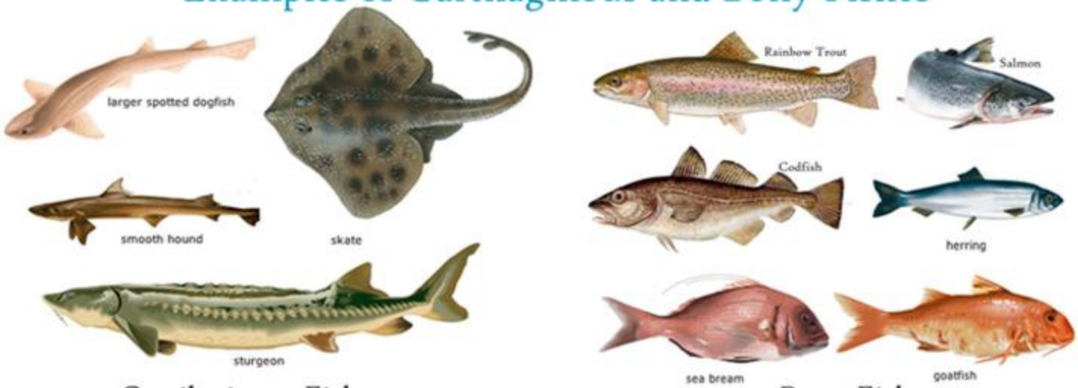






Modern actinopterygian, an example of a gnathostome.

Examples of Cartilaginous and Bony Fishes



Cartilaginous Fishes   Bony Fishes